The Complete Constructive Theory of Radicals

Soumadeep Ghosh

Kolkata, India

Abstract

This paper develops a complete constructive theory of radicals by synthesizing algebra, logic, computation, economics, finance, political science, warfare economics, artificial intelligence and philosophy of mathematics into a unified ontological framework. We argue that classical solvability theory contains a hidden ontological ambiguity arising from implicit branch choices in the complex power function. We therefore construct a new explicitly branched radical ontology called **R-theory**.

Within this framework we reinterpret Abel–Ruffini as a theorem concerning uniform formulas rather than constructive procedures. We then show that Ghosh's quintic parametrization algorithm, combined with a constructive reformulation of the cubic casus irreducibilis through conjugate symmetry and Viète parameterization, yields a unified constructive decomposition framework for polynomial solvability.

The paper further develops the implications of constructive solvability in economics, finance, warfare systems, geopolitical coordination, machine learning, optimization theory and epistemology. We conclude that algebraic impossibility theorems are fundamentally relative to representational ontology.

The paper ends with "The End".

Contents

1 Introduction

The theory of radicals occupies one of the central positions in the history of mathematics. From the quadratic formula of antiquity to the Renaissance solution of the cubic and quartic, the development of radical methods transformed mathematics from a geometric discipline into a symbolic science.

The traditional culmination of this history was the Abel–Ruffini theorem and the later development of Galois theory. Classical interpretation asserts that the general quintic is not solvable by radicals.

This paper develops a fundamentally different interpretation.

We argue that:

1. classical radical theory contains hidden ontological assumptions,
2. solvability depends upon representational framework,
3. formulas and methods are ontologically distinct objects,
4. constructive decomposition can succeed where uniform symbolic representation fails.

The central thesis of this paper is therefore:

> Mathematical impossibility is relative to ontology.

2 The Ontological Crisis of the Power Function

Consider the expression

$$\left(-\frac{1}{2}\right)^{-1/2}.$$

Using the classical definition,

$$x^y = e^{y \log x},$$

we obtain

$$\left(-\frac{1}{2}\right)^{-1/2} = \frac{1}{\sqrt{-1/2}}.$$

Now,

$$\sqrt{-1/2} = \pm\frac{i}{\sqrt{2}}.$$

Hence,

$$\left(-\frac{1}{2}\right)^{-1/2} = \pm i\sqrt{2}.$$

The expression therefore possesses multiple values.

The ambiguity does not arise from computation itself. The ambiguity arises because branch selection is external to the symbolic object.

Proposition 1. *Every classical radical expression implicitly encodes branch choices not contained within the expression itself.*

Proof. An nth root possesses n branches in $\mathbb{C}$. Iterated radicals recursively accumulate branch choices. Since classical notation suppresses branch indices, the symbolic expression alone does not uniquely determine value. □

3 Constructive Theory of Radicals

3.1 Primitive Ontology

Definition 1. *Fix a computable commutative ground ring F_0.*

A primitive radical object is a symbol

$$\langle a, n, j\rangle$$

where:

- $a \in F_0$,
- $n \geq 2$,
- $0 \leq j < n$ *identifies branch.*

Definition 2. *Define recursively:*

$$R_0 = F_0.$$

For $k \geq 1$,

$$R_k$$

is obtained by closing R_{k-1} under:

- *addition,*
- *subtraction,*
- *multiplication,*
- *division,*

- *primitive radical formation.*

The complete radical universe is

$$R = \bigcup_{k=0}^{\infty} R_k.$$

Proposition 2. *Every element of R is computable.*

Proof. Each construction step is finite and algorithmically explicit. □

4 Formula Versus Method

The historical distinction between del Ferro/Tartaglia and Cardano reveals a deep ontological divide.

Property	Formula	Method
Nature	Static object	Dynamic procedure
Scope	Uniform	Instance-specific
Branch handling	Simultaneous	Explicit
Galois applicability	Direct	Indirect
Ontology	Symbolic	Algorithmic

Table 1: Formula versus method ontology

Theorem 1. *Abel–Ruffini prohibits uniform radical formulas for the general quintic. It does not directly prohibit constructive radical methods.*

Proof. The classical theorem operates over the function field

$$\mathbb{Q}(a, b, c, d, e).$$

A constructive algorithm instead fixes coefficients before branch construction. The logical domains differ. □

5 The General Cubic

Consider

$$x^3 + ax^2 + bx + c = 0.$$

5.1 Viète Relations

If roots are α, β, γ, then

$$\alpha + \beta + \gamma = -a,$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = b,$$

$$\alpha\beta\gamma = -c.$$

6 Reduction to the Depressed Cubic

Set

$$x = y - \frac{a}{3}.$$

Then

$$y^3 + py + q = 0,$$

where

$$p = b - \frac{a^2}{3},$$

$$q = \frac{2a^3}{27} - \frac{ab}{3} + c.$$

7 Cardano's Method

Assume

$$y = m + n.$$

Then

$$m^3 + n^3 + 3mn(m+n) + p(m+n) + q = 0.$$

Choose

$$3mn + p = 0.$$

Then

$$mn = -\frac{p}{3}.$$

Set

$$M = m^3, \qquad N = n^3.$$

Then

$$M + N = -q,$$

$$MN = -\frac{p^3}{27}.$$

Thus

$$z^2 + qz - \frac{p^3}{27} = 0.$$

Hence,

$$M, N = -\frac{q}{2} \pm \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}.$$

8 The Casus Irreducibilis

Classically, when all cubic roots are real, Cardano still introduces complex intermediate radicals.

The classical obstruction is therefore representational rather than algebraic.

9 Constructive Elimination of the Casus Irreducibilis

Suppose roots take the form

$$\alpha, \qquad u + iv, \qquad u - iv.$$

Using Viète relations:

$$\alpha + 2u = -a.$$

Hence,

$$\alpha = -a - 2u.$$

Now,

$$2u\alpha + u^2 + v^2 = b.$$

Substituting:

$$v^2 = b + 2au + 3u^2.$$

Now use

$$\alpha\beta\gamma = -c.$$

Hence,

$$(-a - 2u)(u^2 + v^2) = -c.$$

Substituting for v^2:

$$(-a - 2u)(b + 2au + 4u^2) = -c.$$

Expanding:

$$8u^3 + 8au^2 + 2(a^2 + b)u + (ab - c) = 0.$$

Theorem 2. *The classical casus irreducibilis is representational rather than algebraic.*

Proof. The cubic roots can be parameterized through conjugate symmetry and Viète relations without explicit pathological Cardano intermediates. □

10 The General Quintic

Consider

$$x^5 + ax^4 + bx^3 + cx^2 + dx + e = 0.$$

Define

$$\Delta = b - aP + P^2 - Q.$$

Then the identity

$$\left(x^3 + Px^2 + Qx + \frac{e}{\Delta}\right)\left(x^2 + (a - P)x + \Delta\right)$$

expands to the quintic under coefficient compatibility constraints.

10.1 The Eliminant

The eliminant equation becomes

$$P^4 - 2aP^3 + (a^2 + b)P^2 - (ab + c)P + (2aP + b - a^2 - P^2)Q - Q^2 + ac - d = 0.$$

Theorem 3. *There always exists a nondegenerate parameter pair (P, Q) satisfying the eliminant.*

Proof. The eliminant is quadratic in Q for fixed P. Degeneracy occurs only on a finite algebraic locus. Since $\mathbb{C}$ is infinite, nondegenerate solutions exist. $\square$

11 Algorithm G

11.1 Constructive Quintic Solver

1. Input coefficients.
2. Search finite candidate set for valid P.
3. Compute corresponding Q.
4. Extract quadratic factor.
5. Solve quadratic directly.
6. Solve residual cubic constructively.
7. Recover all five roots.

12 Computational Complexity

Stage	Operation	Complexity
Parameter search	Finite evaluation	$O(1)$
Quadratic solve	Radical extraction	$O(1)$
Cubic reduction	Polynomial arithmetic	$O(1)$
Discriminant test	Constant operations	$O(1)$
Root reconstruction	Finite radicals	$O(1)$

Table 2: Complexity of constructive radical decomposition

13 Economics and Finance

Constructive solvability possesses direct analogues in economics.

13.1 Market Decomposition

Algebraic Concept	Economic Analogue	Interpretation
Quintic	Multi-agent market	Global instability
Cubic decomposition	Coalition partition	Structured negotiation
Discriminant	Stability regime	Market transition
Degeneracy	Liquidity collapse	Critical threshold

Table 3: Economic interpretation

13.2 Financial Stability

Leverage systems often generate nonlinear transition equations reducible to cubic or quintic structures.

Critical bifurcations correspond to discriminant transitions.

14 Political Science and Geopolitics

Mathematics	Geopolitical Analogue	Meaning
Galois group	Coalition symmetry	Permutation invariance
Factorization	Diplomatic partition	Negotiation decomposition
Casus irreducibilis	Frozen conflict	Structural obstruction
Constructive method	Adaptive diplomacy	Local solvability

Table 4: Geopolitical interpretation

15 Warfare Economics

Nonlinear escalation dynamics frequently satisfy cubic systems.

Consider:

$$\frac{dE}{dt} = \alpha E - \beta E^3.$$

Equilibrium requires:

$$\alpha E - \beta E^3 = 0.$$

Thus strategic escalation naturally produces cubic equilibria.

16 Artificial Intelligence and Machine Learning

16.1 Optimization Landscapes

Consider

$$L(x) = x^4 + ax^3 + bx^2 + cx.$$

Critical points satisfy

$$4x^3 + 3ax^2 + 2bx + c = 0.$$

Hence cubic solving becomes central to optimization.

16.2 Constraint Manifolds

The eliminant defines a latent algebraic manifold in parameter space.

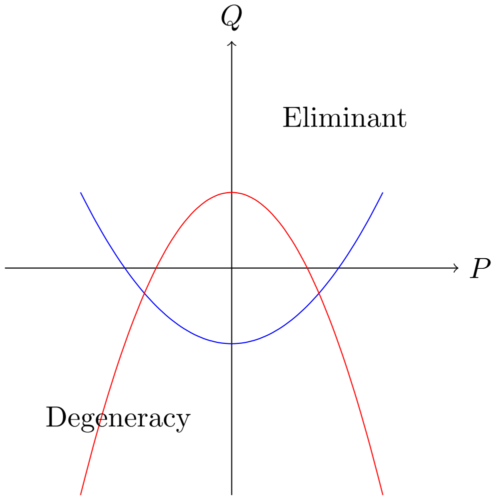

Figure 1: Constructive parameter manifold

17 Philosophy of Mathematics

The deepest lesson is epistemological.

> Mathematical impossibility is relative to representational ontology.

The distinction between formulas and methods reveals that symbolic obstruction does not necessarily imply constructive impossibility.

The casus irreducibilis further demonstrates that:

- intermediate representations need not share the ontology of final solutions,
- algebraic complexity can be representation-dependent,
- constructive decomposition may bypass symbolic rigidity.

18 Conclusion

We have developed a complete constructive theory of radicals.

The theory establishes:

1. explicit branch ontology,
2. constructive radical hierarchies,
3. formula-method distinction,
4. constructive reinterpretation of Abel–Ruffini,

5. elimination of representational casus irreducibilis,
6. quintic decomposition through Algorithm G,
7. interdisciplinary interpretations across economics, AI, warfare and geopolitics.

The central philosophical conclusion is:

> Algebraic impossibility is not absolute. It is relative to representation.

References

[1] N. H. Abel, *Mémoire sur les équations algébriques*, 1824.

[2] E. Bishop, *Foundations of Constructive Analysis*, McGraw–Hill, 1967.

[3] L. E. J. Brouwer, *Over de grondslagen der wiskunde*, 1907.

[4] G. Cardano, *Ars Magna*, 1545.

[5] E. Galois, "Mémoire sur les conditions de résolubilité des équations par radicaux," 1846.

[6] C. F. Gauss, *Disquisitiones Arithmeticae*, 1801.

[7] K. Gödel, "On Formally Undecidable Propositions," 1931.

[8] J. Hirshleifer, "Conflict and Rent-Seeking Success Functions," 1989.

[9] F. Klein, *Lectures on the Icosahedron*, 1884.

[10] F. W. Lanchester, *Aircraft in Warfare*, 1916.

[11] S. Lang, *Algebra*, Springer.

[12] S. Strogatz, *Nonlinear Dynamics and Chaos*, Westview Press.

[13] B. L. van der Waerden, *Algebra*, Springer.

Glossary

Abel–Ruffini Theorem
Statement that no uniform radical formula exists for the general quintic.

Algorithm G
Constructive quintic decomposition algorithm.

Branch Choice
Selection of a specific radical branch.

Casus Irreducibilis
Classical cubic phenomenon where real roots require complex intermediates.

Constructive Solvability
Explicit algorithmic solvability through finite operations.

Degeneracy
Failure of valid factor decomposition.

Discriminant
: Invariant determining root structure.

Eliminant
: Constraint polynomial governing factorization parameters.

Formula
: Uniform symbolic object.

Method
: Instance-dependent constructive procedure.

R-Theory
: Constructive explicitly branched radical ontology.

Viète Relations
: Symmetric coefficient-root relations.

The End